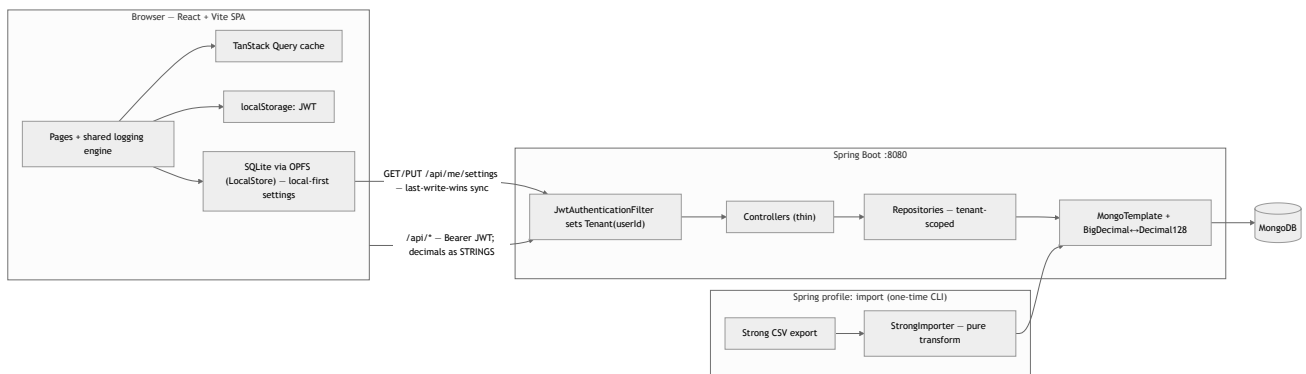


Workout Logger — diagrams

Structural and behavioural diagrams (Mermaid, renders on GitHub). 17 validated diagrams. See `DESIGN.md` for the authoritative architecture record and `CLAUDE.md` for invariants. Diagrams reflect the current code.

Structural

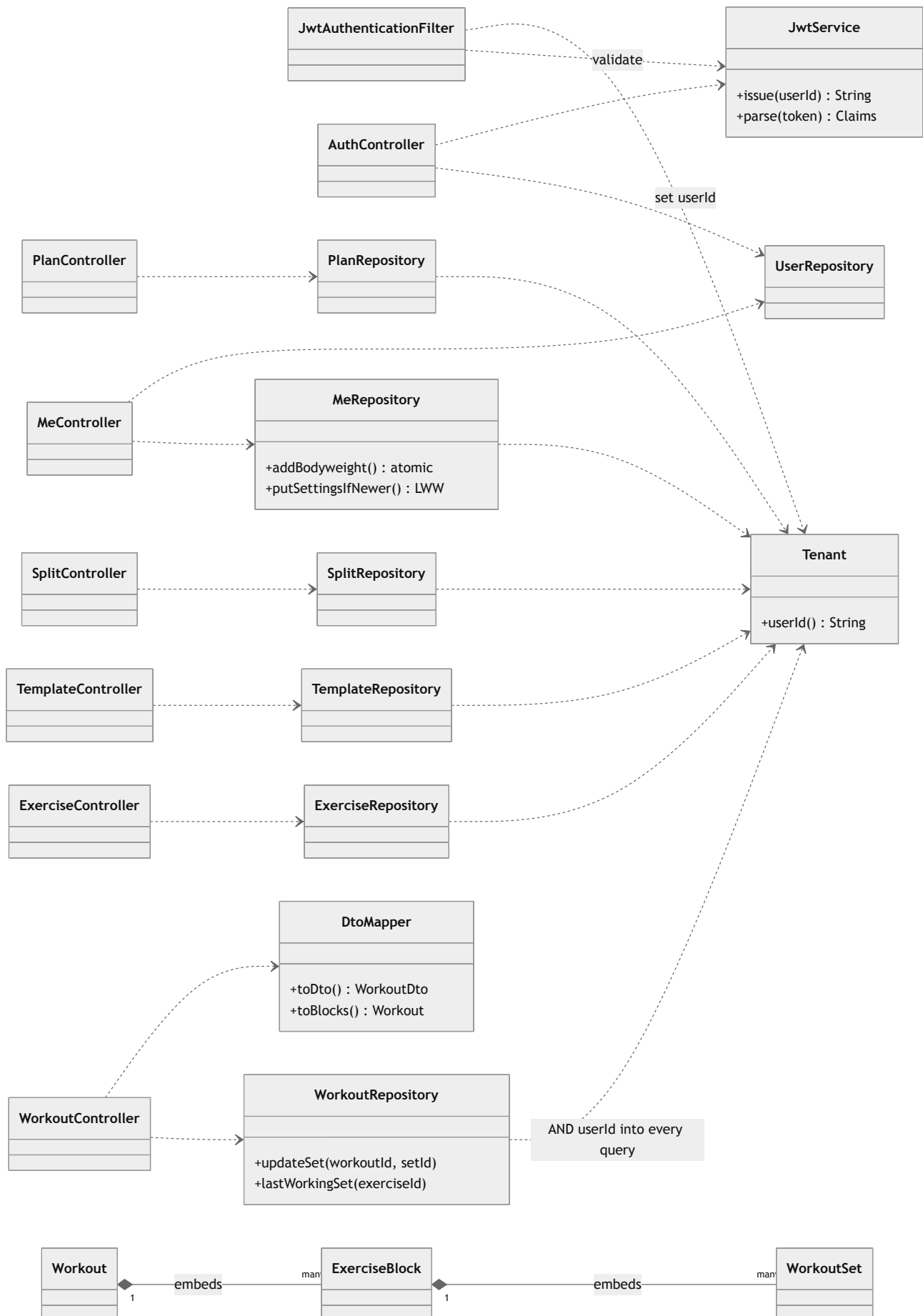
1. System architecture (components & runtime)



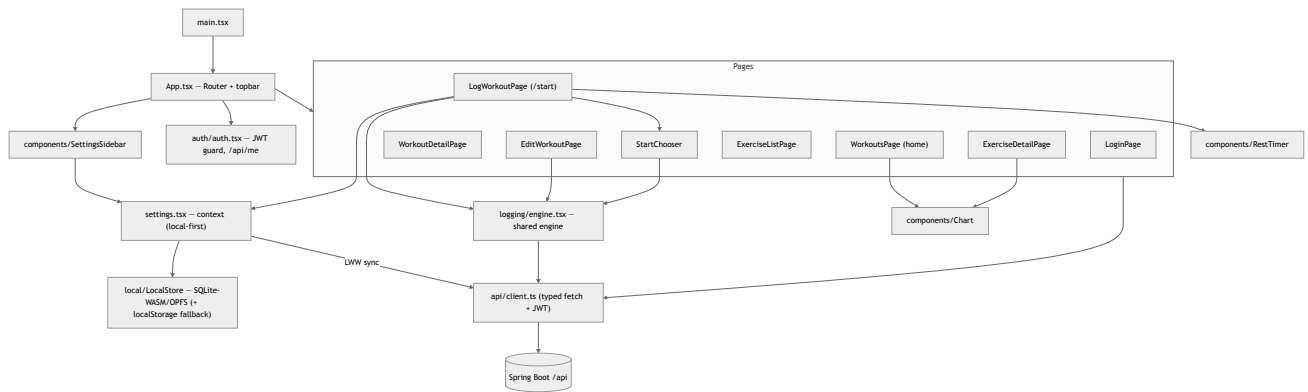
2. Data model (collections, embedding & references)

A **workout session** is one document that embeds `exercises[]`, each embedding `sets[]`. Everything else relates by id reference (many-to-many), not joins. Every collection is scoped by `userId`.

3. Backend layers (request path + tenant isolation)

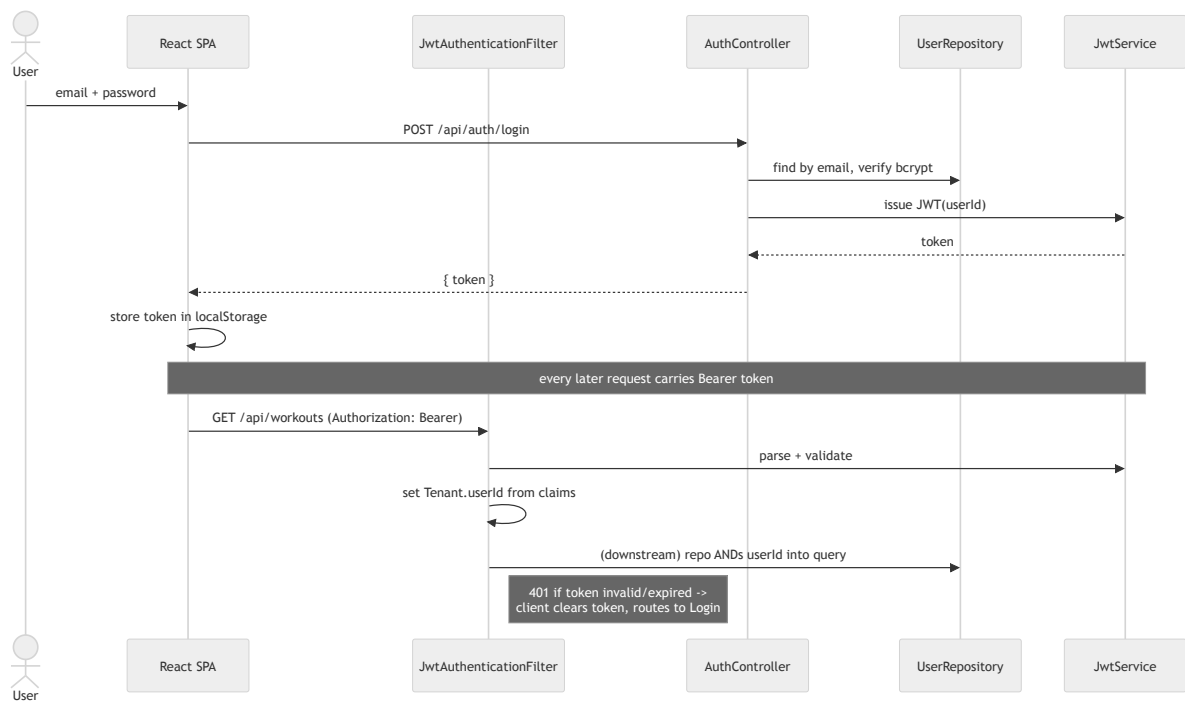


4. Frontend modules

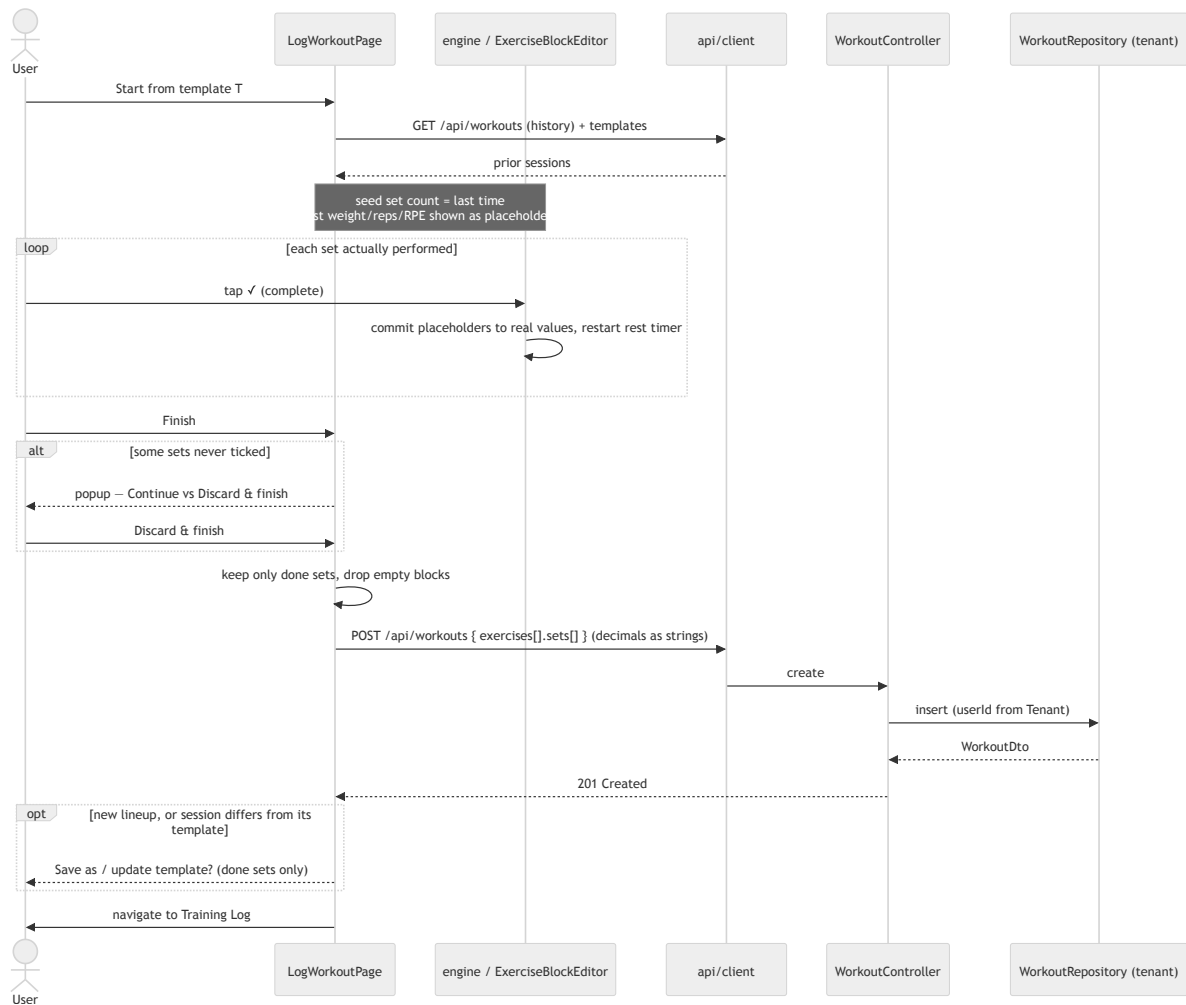


Behavioural

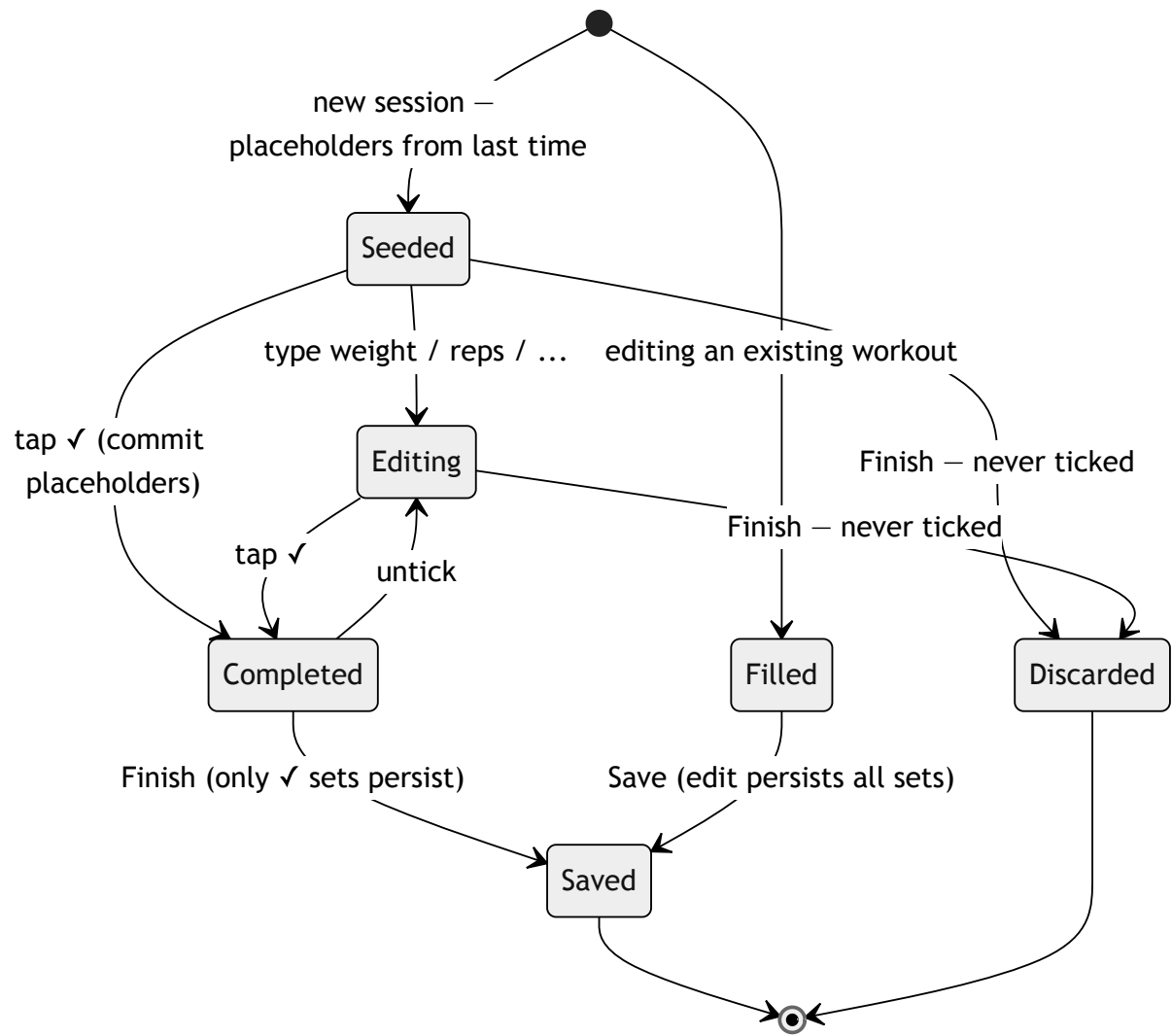
5. Authentication & per-request tenant isolation



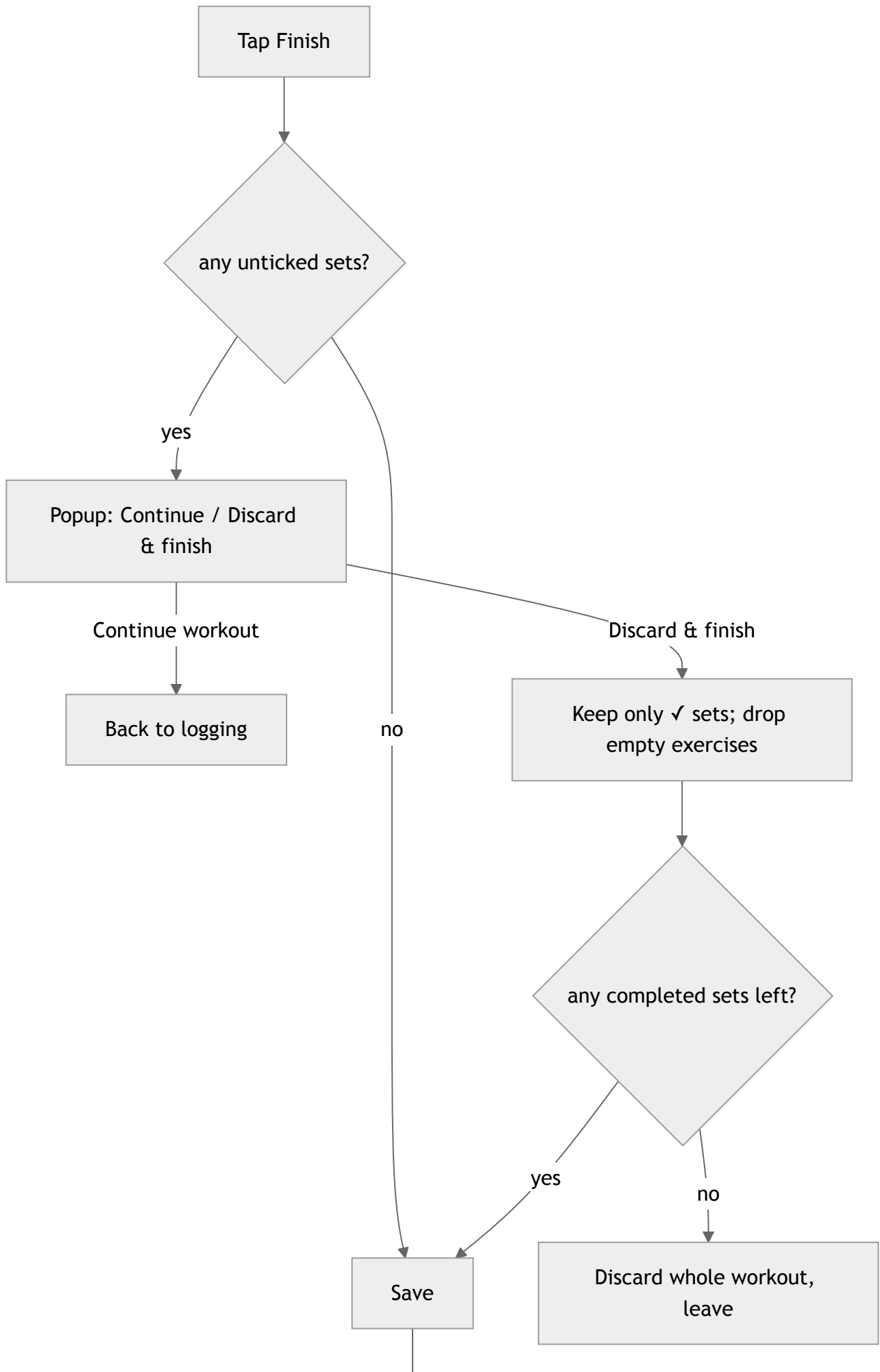
6. Logging a workout (start → complete → finish → save)

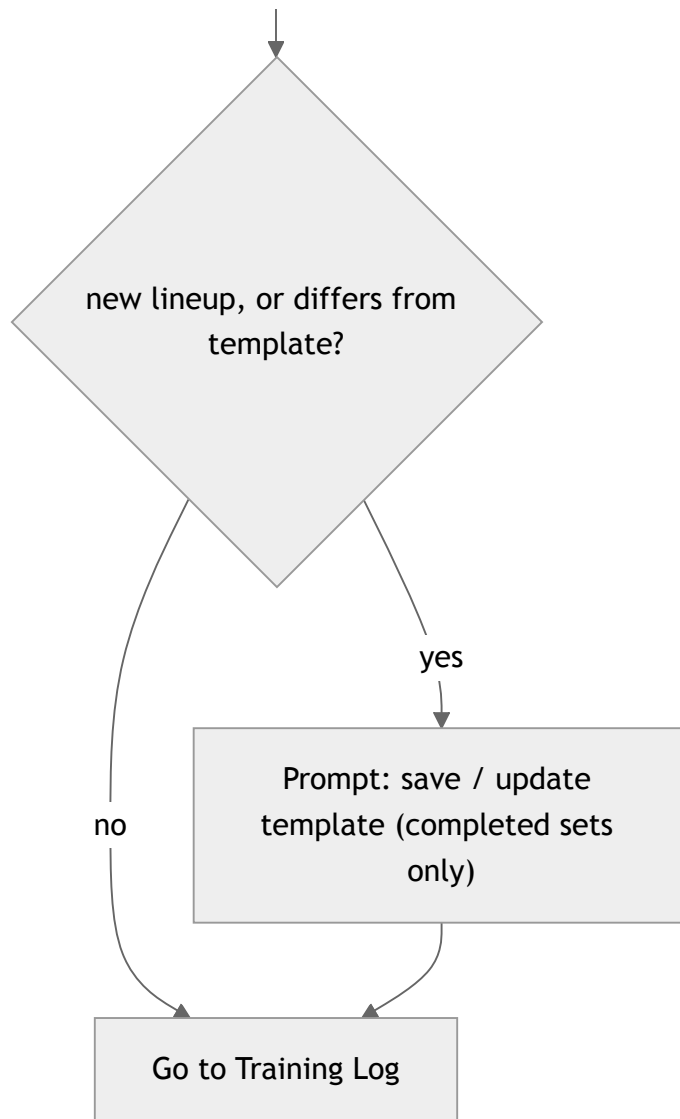


7. A set's lifecycle (placeholder → completed → saved/discarded)

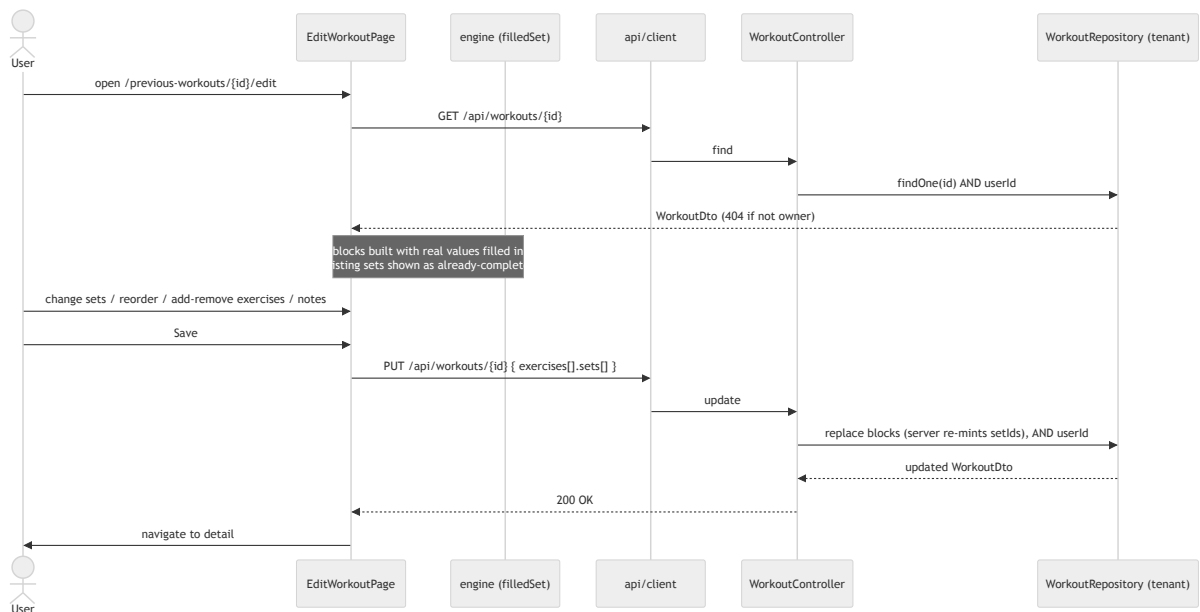


8. Finish-workout decision (discarding unfinished sets)

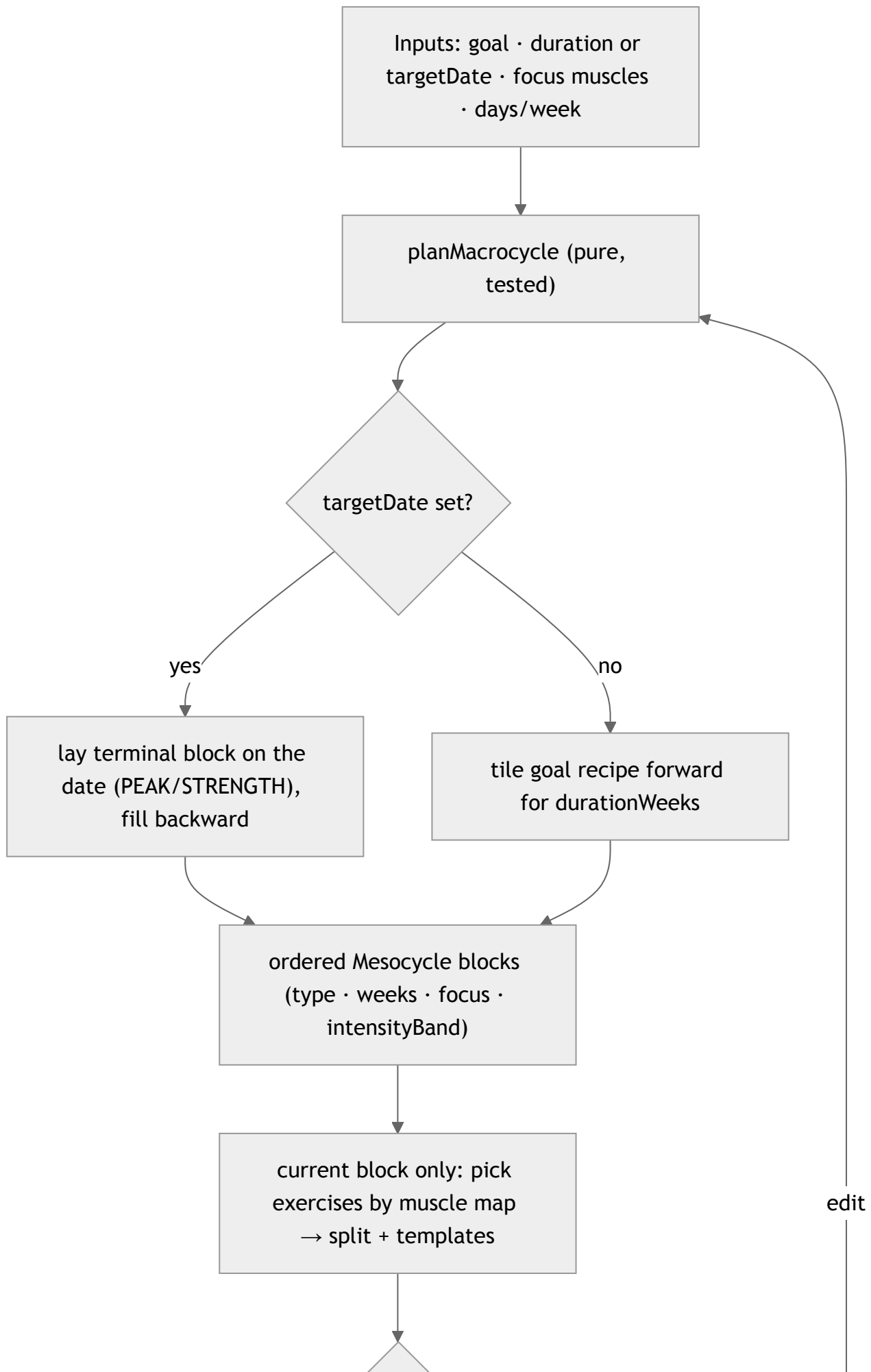


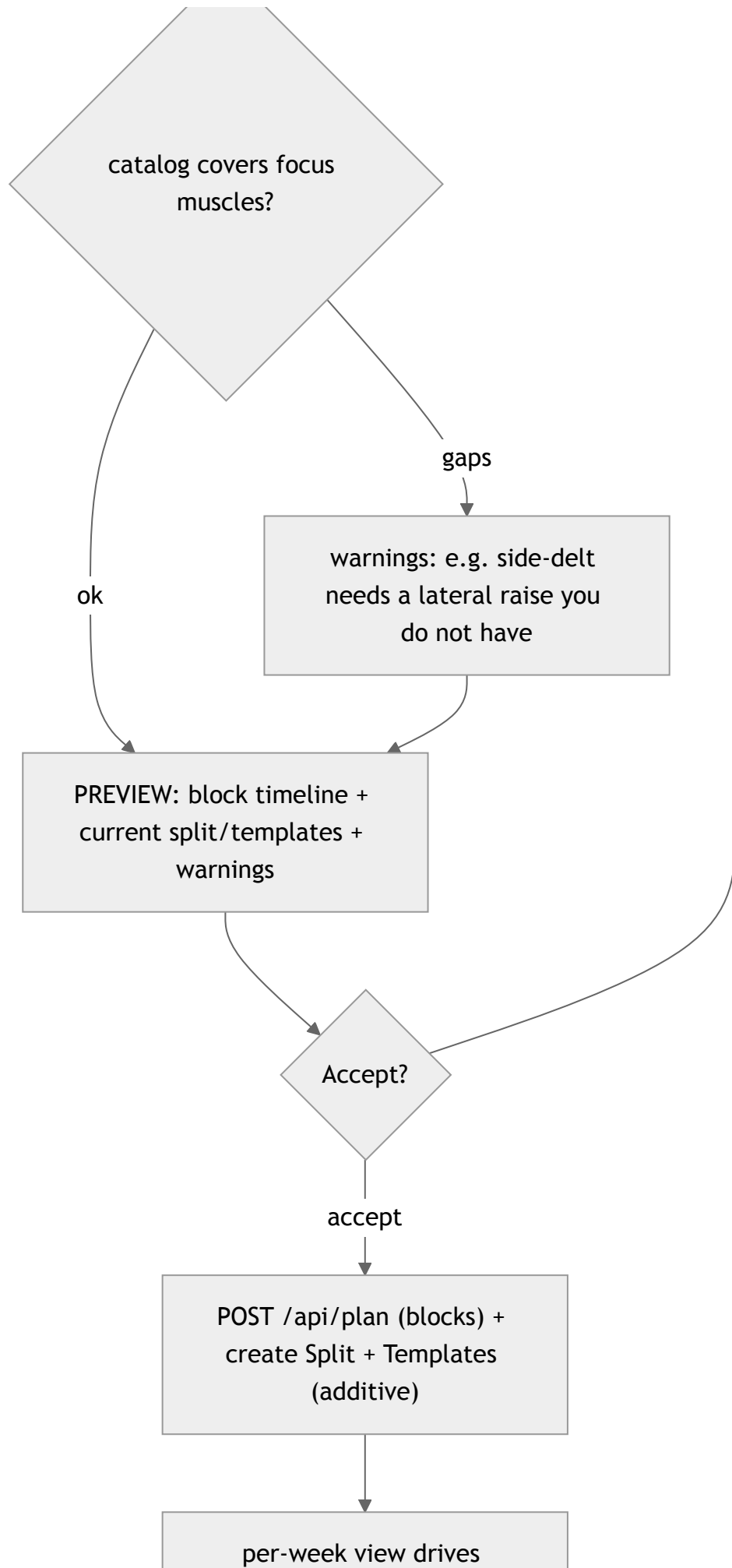


9. Editing a completed workout



10. Macrocycle planner (Layer 4)

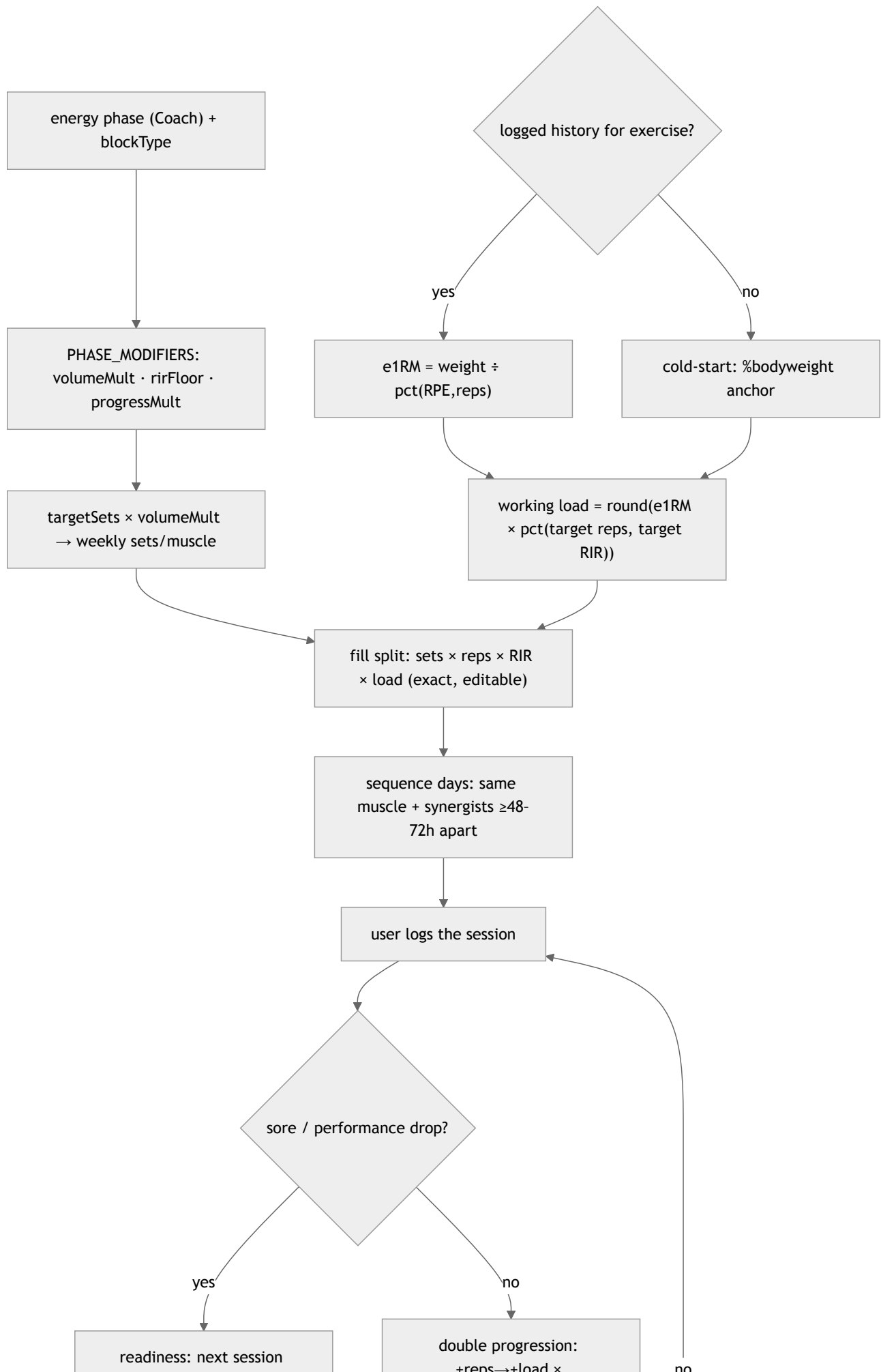


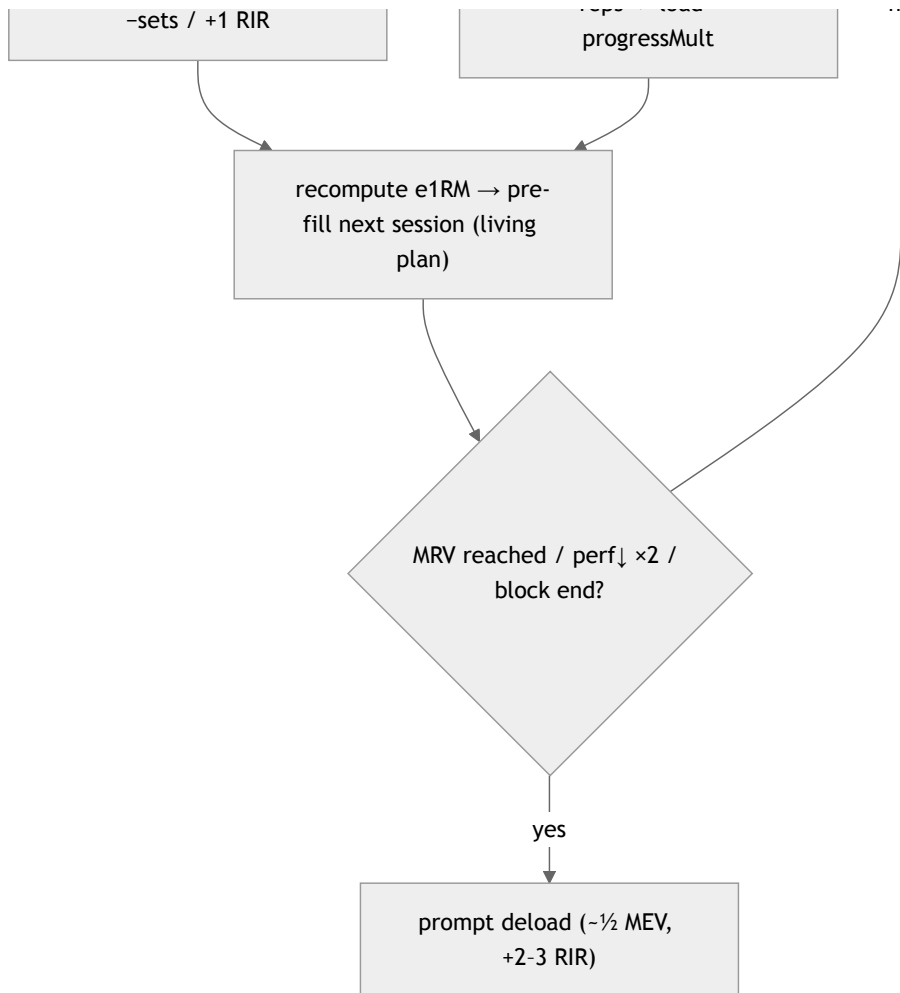


logging; advance through
microcycles

blockType (volume band + reps) and energy phase (deficit-trim) are orthogonal axes; accept creates, never mutates; only the current block's training is materialized — distal blocks stay as intent.

11. Prescription, recovery & autoregulation (Layer 5)

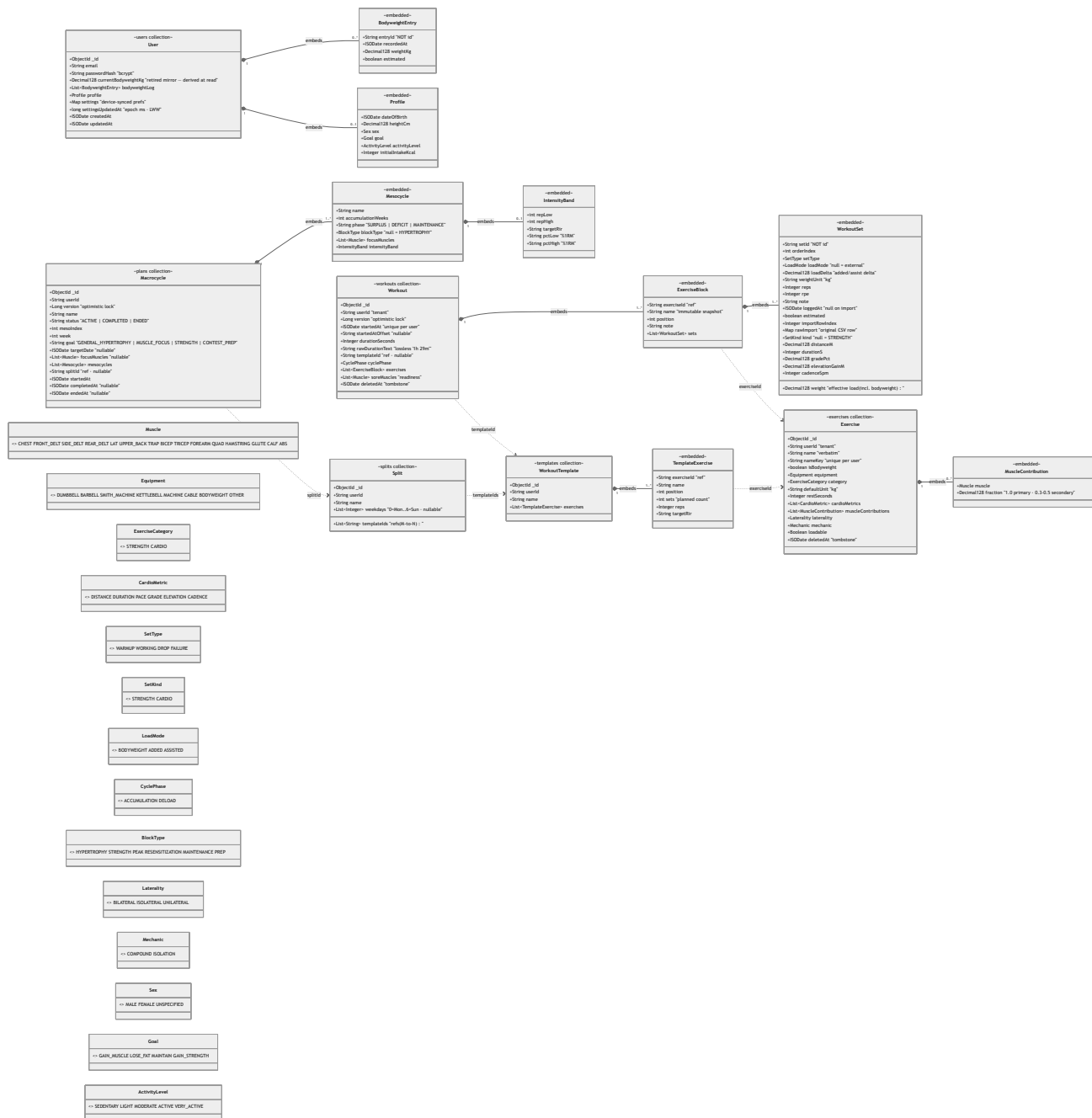




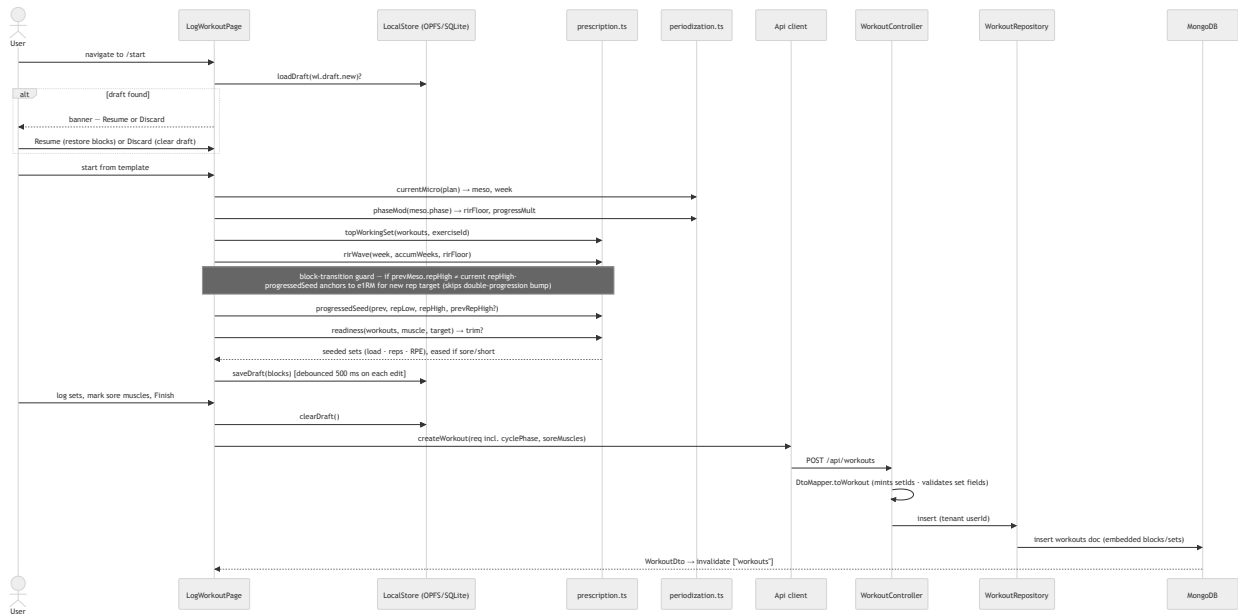
Energy phase scales volume/intensity/progression; numbers seed from logged e1RM (else %BW cold-start); recovery spacing + readiness keep a muscle from being trained fatigued; everything stays an editable preview.

12. Domain model — full class diagram (every field, shown as it is STORED)

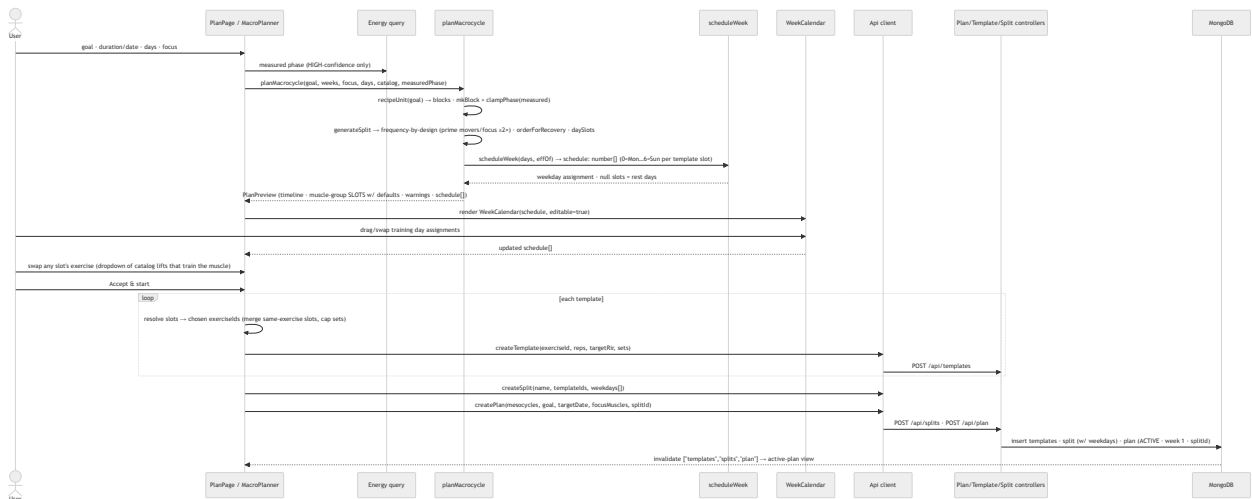
Types are the **Mongo storage types**, not the Java types: an @Id is an ObjectId ; a cross-reference (userId , templateId , exerciseId , splitId) is the plain hex **String**; a weight is Decimal128 (serialized as a **string** on the wire); a timestamp is ISODate . Boxes tagged <<embedded>> have no collection of their own — they live inside their parent document. The six collection roots are tagged with their collection name.



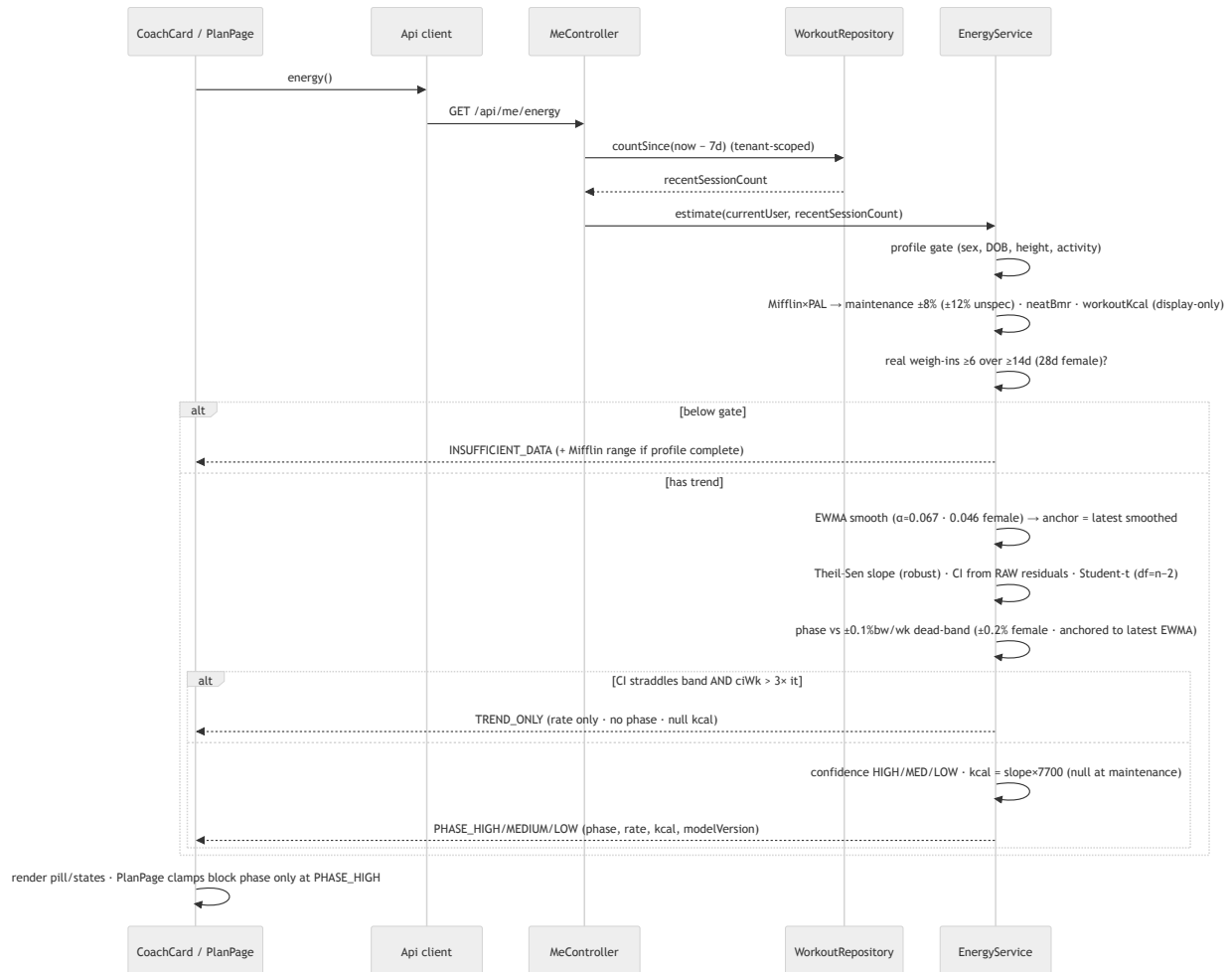
13. Sequence — log a planned session (the living plan)



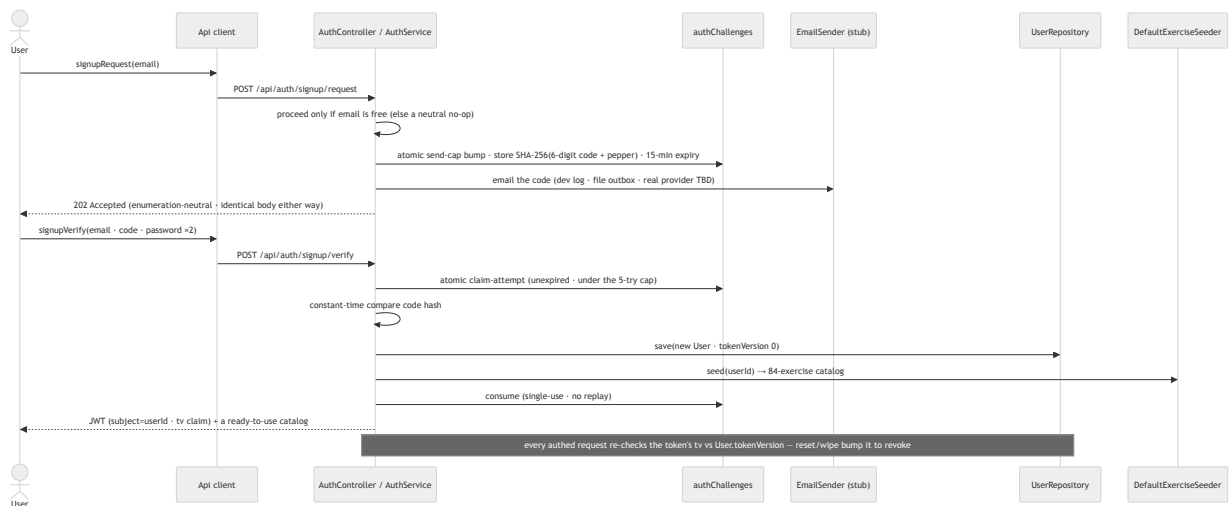
14. Sequence — build & accept a macrocycle plan



15. Sequence — energy "Coach" estimate (read-time, gated)



16. Sequence — verified sign-up (code) → account + default-catalog seeding



17. Sequence — plan completion + history

